

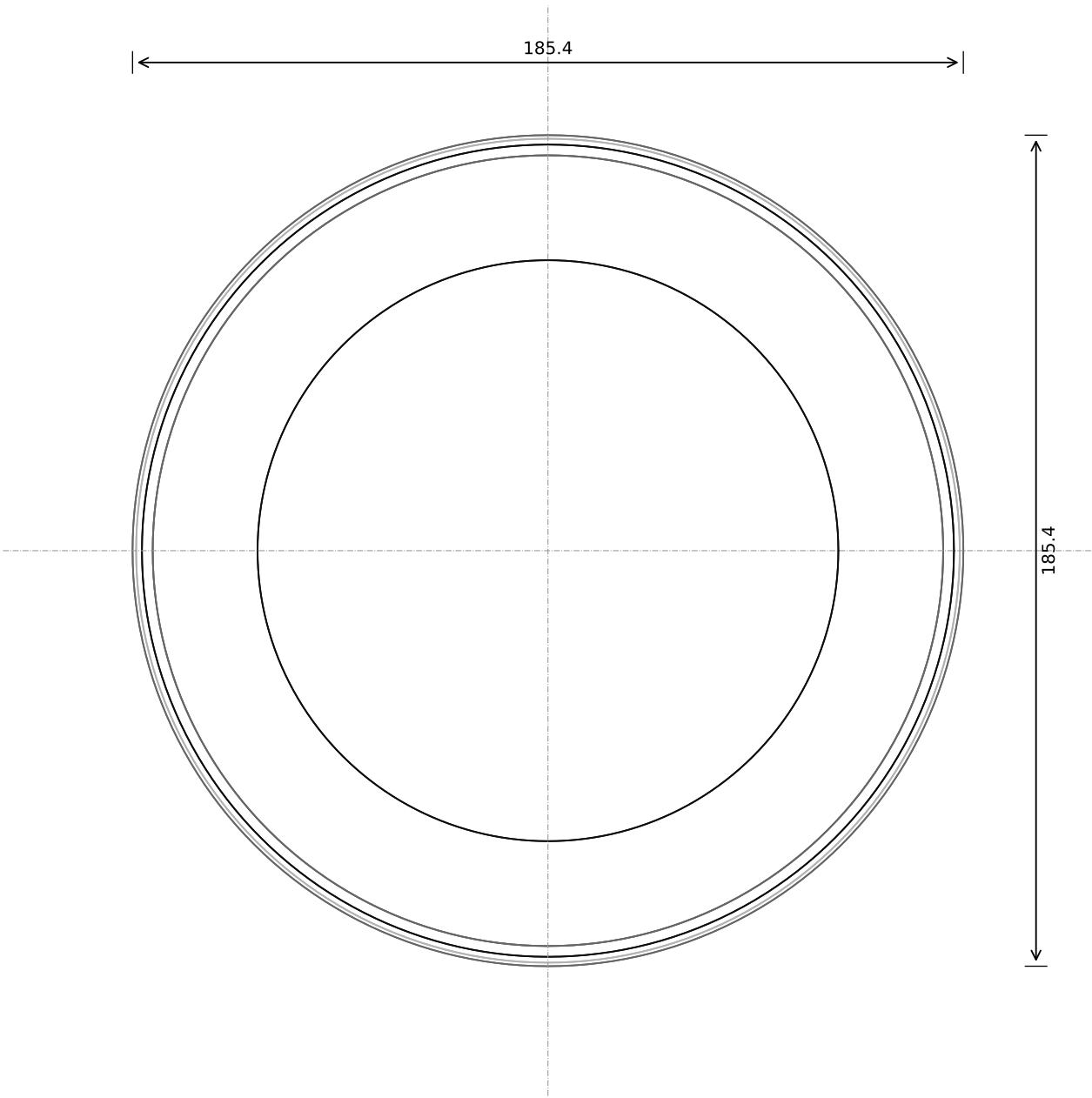
the 60 — v16 — sheet 0: the ASSUMED numbers (every one of these must be verified on the real part before the drawing is trusted)

2026-09-08. Stack: plate 8.0, adapter 14.78-16.38 (socket to 23.68), code ring under the crown at 31.53, seat 26.0, glass 26.50-28.48, lens to 30.98, knob top 34.38, overall 35.88 with the pad. Ø185.

1. cover lens: Ø140.0 x 2.5	specified separately; carrier rim and lip overlap designed to it
2. bonding tape: 0.5	double-sided foam tape between the seat and the glass edge (RULING 5 Sep: glue allowed for the display)
3. panel back component area: 60.0 x 68.36 x 1.5 at -14.0	position from the drawing's back view
4. Active Cooler position: from Pi x 2.5, fan from x 43.0	no CAD body; footprint and fan position from photographs
5. cooler base above the board: 2.8	SoC 2.4 + thermal pad
6. fan intake plenum: 8.0	minimum free height above the blower
7. micro-HDMI receptacles: x 26.0/39.5	not in the vendor STEP; drawn as envelopes
8. HDMI ribbon plugs: 12x8x5 micro end, 21x12x6 A end	flat FPC HDMI cable, 200 mm
9. adapter board thickness and part heights: 1.6; parts 2.5, underside 1.0, flex connector 2.0	outline, holes, socket size and position PUBLISHED (datasheet drawing)
10. display connect board: 30.0 x 46.0 x 1.2, connectors 2.0	part of the adapter kit; size scaled from its photograph, hole positions assumed at the corners
11. Pi standoffs and washers: M2.5 x 12.0 hex, 0.5 washers	the adapter stack: closing-plate land / washer / Pi / standoff / adapter
12. audio / touch boards: envelopes	electrical layout not done
13. bought breakouts: TMC6300-BOB 25.4 x 20.3; DRV2605L 20.0 x 15.0; 74AHCT125 20.0 x 15.0; hole patterns	the vendors' pages where published; hole positions and part heights assumed
14. microcontroller: ESP32-S3-Zero class 25.0 x 18.0, USB-C 3.2 tall	RULING pending (question 33): a Pico 2 does not fit the tray
15. converter height: 7.6	Pololu D24V90F5: outline published, the inductor's height not
16. V-groove bearing: V623ZZ: V 1.2 deep, 0.4 root flat, 90.0 deg	the ridge is derived from these - measure Ryan's listing and re-run
17. block spring: Ø4.0 x 16.0 free, 0.8 N/mm	about 4 N seated; the bench sets it (the band slip limit is 120-160 mN.m at 4 N)
18. light-sensor breakout: sensor 3.0 from the edge	Adafruit 4162, from Ryan's photograph
19. panel jack body: Ø9.0 x 14.0	the listing's drawing to fetch
20. Mill-Max spring-pin block: 5.1 x 2.6 x 3.0, pins Ø1.5 x 2.2 compressed	867-22-002-70-501010; datasheet to fetch
21. diffuser gasket: 0.3	closed-cell foam ring under the lip
22. halo liner: 0.4 floor, 3.4 wall	white PETG; a white coating on the ring is the alternative (throw 4.0 instead of 3.6)
23. barrel jack body: 14.4 x 9.0 x 11.0 on a vertical board	PJ-063AH; CAD body to fetch
24. right-angle barrel plug: Ø8.5 body	measure the brick's plug
25. MOTOR_BASE_OD / MOTOR_BASE_H: 30.0 / 4.3	carried from v9
26. MOTOR_BAND_T: 0.6	carried from v9
27. servo pushrod: 3 mm bridge to the tab	carried from v9; servo on the plate in a frame
28. LED strip: 4.0 wide x 1.2 thick, face down	a 4 mm 2020 strip (HALO-OPTICS); the channel is 4.4 wide - a 5 mm strip needs the wall 1.0 further in
29. blower inlet and outlet: inlet Ø24.0 in the face, outlet 21.0 x 7.0 from 1.5 up	from the datasheet drawing (outline, holes and leads PUBLISHED)
30. blower gaskets: 0.3 closed-cell foam	under the inlet face and the hood
31. closing-plate gasket: 0.2 die-cut	in a groove of the recess floor, on every sealing wall
32. Pi 5 fan connector current rating: 0.13 A drawn (0.20 max)	the connector sits in the vendor STEP; its rating is not published
33. encoder shim family: [0.7, 0.8, 0.9, 1.0, 1.1]	printed; the height is chosen on the bench

Build notes in V16.md. The sheets that follow: one per made part.

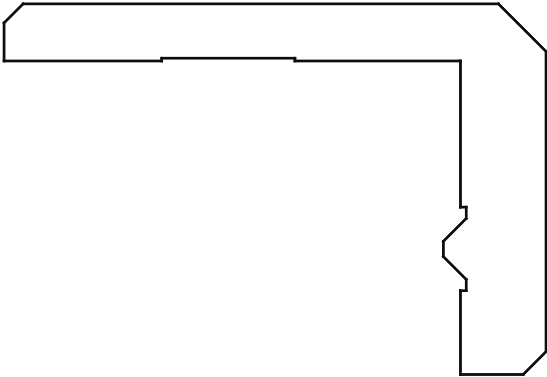
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 63 to 95, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- outside Ø185.4, bore Ø176.4 (0.45 to the panel's chin corners); wall 4.5
- top rim: lip inner edge Ø128.4, 1.0 inner chamfer, 25 flat, 2.5 outer chamfer (RULING 6 Sep: rim >= 20); lip 3.0 thick, top at z 34.38
- outer chamfer 2.5 x 45° smooth; inner chamfer 1.0; skirt chamfer 1.2
- v16 V-RIDGE on the bore: relief band r 88.50 z 19.3-23.7; the ridge 90° flanks from its base (3.2 wide, z 19.9-23.1) to the crest r 87.30 (0.8 flat) - three V623ZZ V-groove bearings straddle it
- code RING recess 0.15 deep in the crown's underside, r 72.5-79.5 (v13: the encoder reads the crown, not the bore - the bore is plain apart from the V-groove)
- diamond knurl 82 starts x 2, 1.0 deep, helix 36.9°, 3 whole rows z 16.9-31.1; skirt bottom z 14.9
- wall under the knurl root 3.5

the 60 — v16

Knob

Cadrane · sheet 1 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

Material: PETG (prototype) / CNC 6082-T6, hard anodised (production)
Process: print top face down, 0.16 mm; no supports
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: knob_body

Overall 185.4 x 185.4 x 19.5
Volume 81.96 cm³
Scale: see each view (fit)

Technical drawing of a circular mechanical component, likely a cross-section of a wheel or a similar structure. The drawing shows a circular outer rim with a central hub area. The outer rim is divided into segments by radial lines. The central hub area contains several small circular features, possibly holes or mounting points. The drawing includes dimension lines indicating a diameter of 185.4 and a width of 185.4.

SECTION B-B (side, plane $x = 0$; y across, z up)

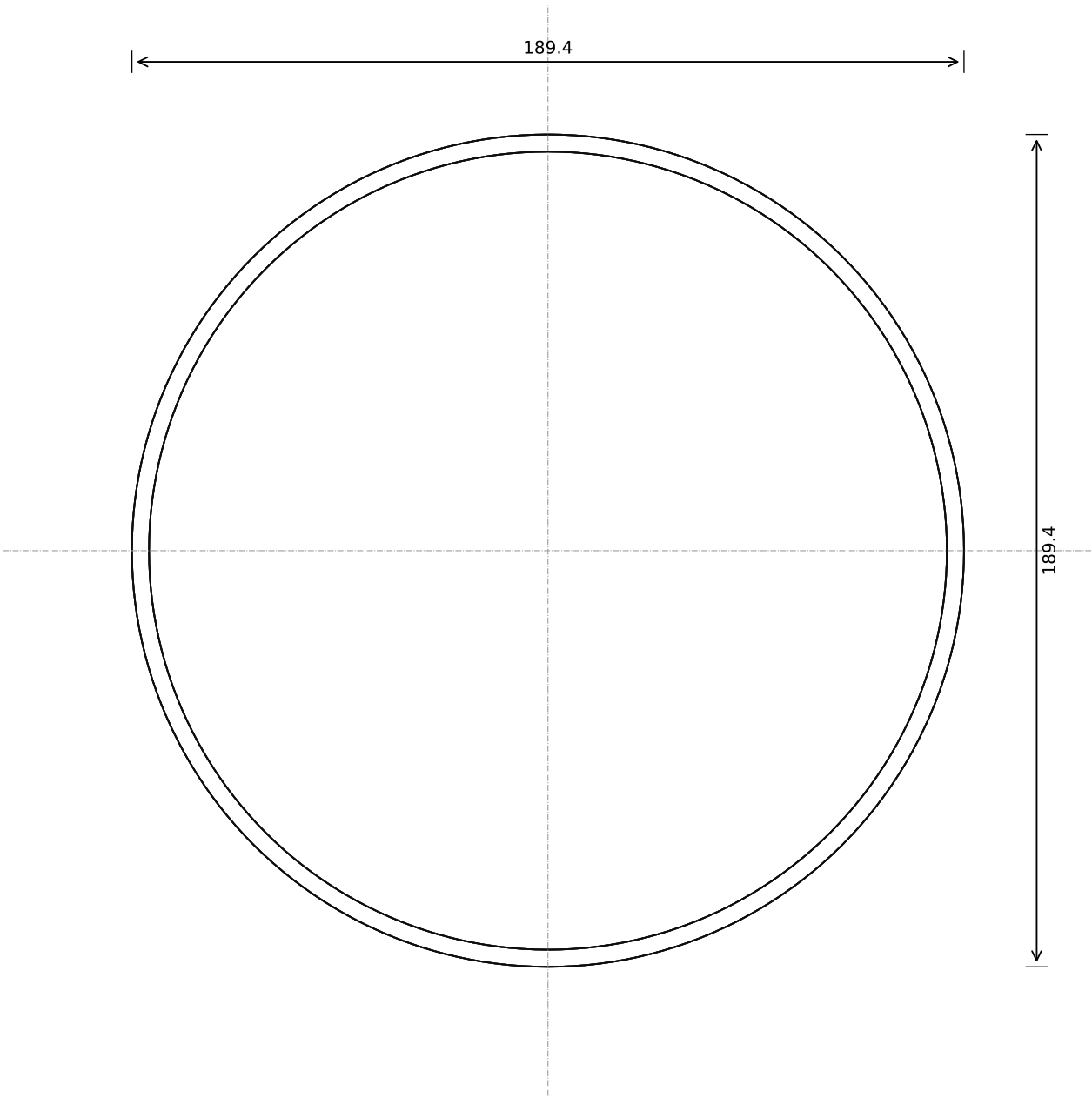
This side view shows the internal structure of the bracket and its connection to the base plate. The bracket's internal features, including the vertical support and the angled transition, are visible. The base plate is shown as a thick horizontal line at the bottom, with a small rectangular feature on its right side.

1. wall in TWO radii: r 84.80-86.40 from the plate to z 13.8, a 50° ramp inward to r 82.80-84.40 at z 16.2, then up to the seat (the knob is fitted 2.4 off-centre and its far-side is 1.0 from the plate)
2. seat flange r 62.0-84.40, z 24.0-26.0: its top face is FLAT with 0.3 chamfers on both edges; three Ø2.0 H7 through-holes (ream) at [70, 210, 330]°, r 67.25 for the display's cable
3. HALO (HALO-OPTICS): the strip's ledge r 84.9-90.50, z 13.20-14.3 (the 4.0 mm strip stuck face DOWN under it, r 86.6-90.6, 55 LEDs, 2.64 A full white, firing at the white line)
4. two FIXED wheel posts Ø8.0 at r 82.3, az [30, 150], to z 19.0, Ø2.9 press holes for the Ø3 m6 x 10 pins; the SPRUNG wheel's channel at 270°: walls 2.0 at t ±4.2-6.2 from the plate
5. pillars Ø7 at r 80.80, az [160, 240, 328], to z 16.0; M2.5 inserts (Ø3.5 x 4.0) from below (the plate screws) - nothing screws in from above
6. encoder at 335.0°, r 76: two radial slots through the flange at t ±4.75 over M2 inserts in Ø5.0 bosses under it; LRA pad at 258.0° (r 81.6 to the wall) with the contact carrier
7. flex slot 32 x 2.5 through the seat at 0°, offset -4; 102 vent slits 1.0 x 4.5 at z 18.5-23.0, 2.5° grid; motor relief at 90° (the bell's window 1.0 clear, the collar's 2.0, open up to the plate)

Cadrane · sheet 2 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

bowel pins, each in a Ø5.0 boss 4.0 under the flange er 3.6 below): the lip r 90.50-92.70, z 13.5-14.3 clamps the opal ring through a 0.3 Material: PETG (prototype) / PA12 (production) 58.0 to the wall, z 8.0-14.0, roof z 13.2-14.0 with a 3.2 slot for the pin (travel 4.0 in / 0.5 out), end wall r 58.0-60.0 with a Ø2.4 hole at z 10.5 for the relea Process: v16: print INVERTED - the seat flange's top face on the bed, 0.20 1's min. beside its flange tail at 252.5° (two ribs from r 74.5 to 0.4 groove each, 45° to the flange); halo tail notch at 325.0°, the far-end feed's at 190.0° static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	gasket, 0.3 radial clearance Part: internal structure Overall Ø5.4 and Ø5.4 x 16.0 Volume 9.55 cm³ Scale: see each view (fit)
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PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



5.7

DETAIL C (section A-A, x 65 to 97, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- r 90.8-94.7 (3.9 wall), z 7.5-13.2 (5.7 tall); vertical outer face standing 2.0 proud of the knob and plate (Ø189.4)
- 0.3 chamfer on all four edges
- stands in the rim ring's 0.5 rebate (r 90.50 outward, floor z 7.5) on the 0.8 shelf; 0.3 radial clearance to the rebate's wall and the ledge; clamped axially by the lip through a

the 60 — v16

Halo ring (opal acrylic)

Cadrane · sheet 3 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

0.3 foam gasket - never hard between two machined faces

Material: cast opal acrylic, 6 mm sheet (Perspex Opal 030 ~68 % / PLEXIGLASS® Opal 030 ~83 % class), CNC routed, annealed 80 °C 2

Process: machined, not printed (HALO-OPTICS \$6)

Overall 189.4 x 189.4 x 5.7

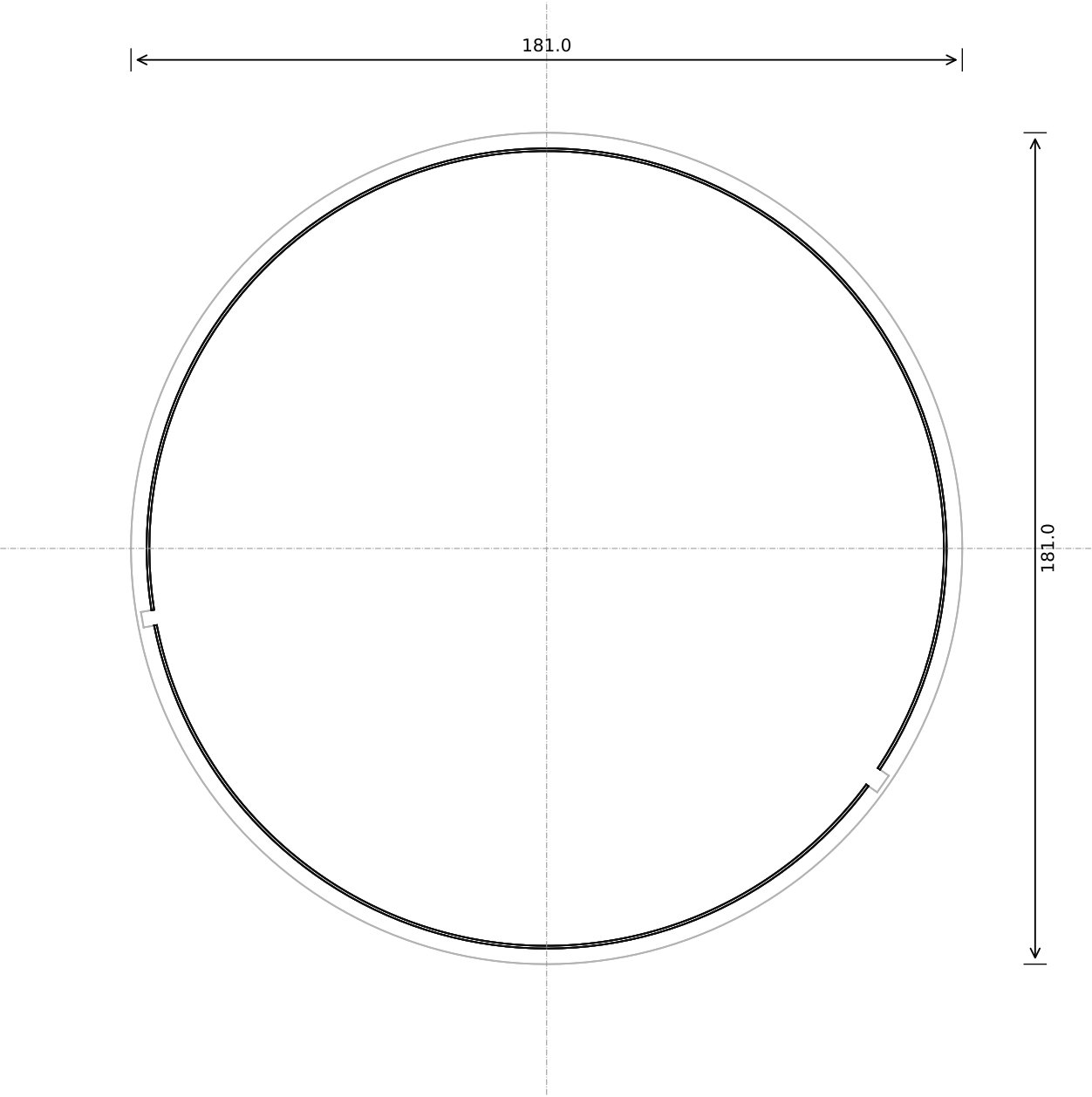
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,

Volume 12.85 cm³

static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



3.40

DETAIL C (section A-A, x 61 to 93, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. L-section ring: floor r 86.5-90.50 x 0.4 on the rim ring's top, wall r 86.5-87.1 up to z 11.4 against the structure's wall (0.6 under the strip); cut at the two tail notches
- 2. HALO-OPTICS §9: the floor and walls must be white and diffuse - bare stainless is specular

the 60 — v16

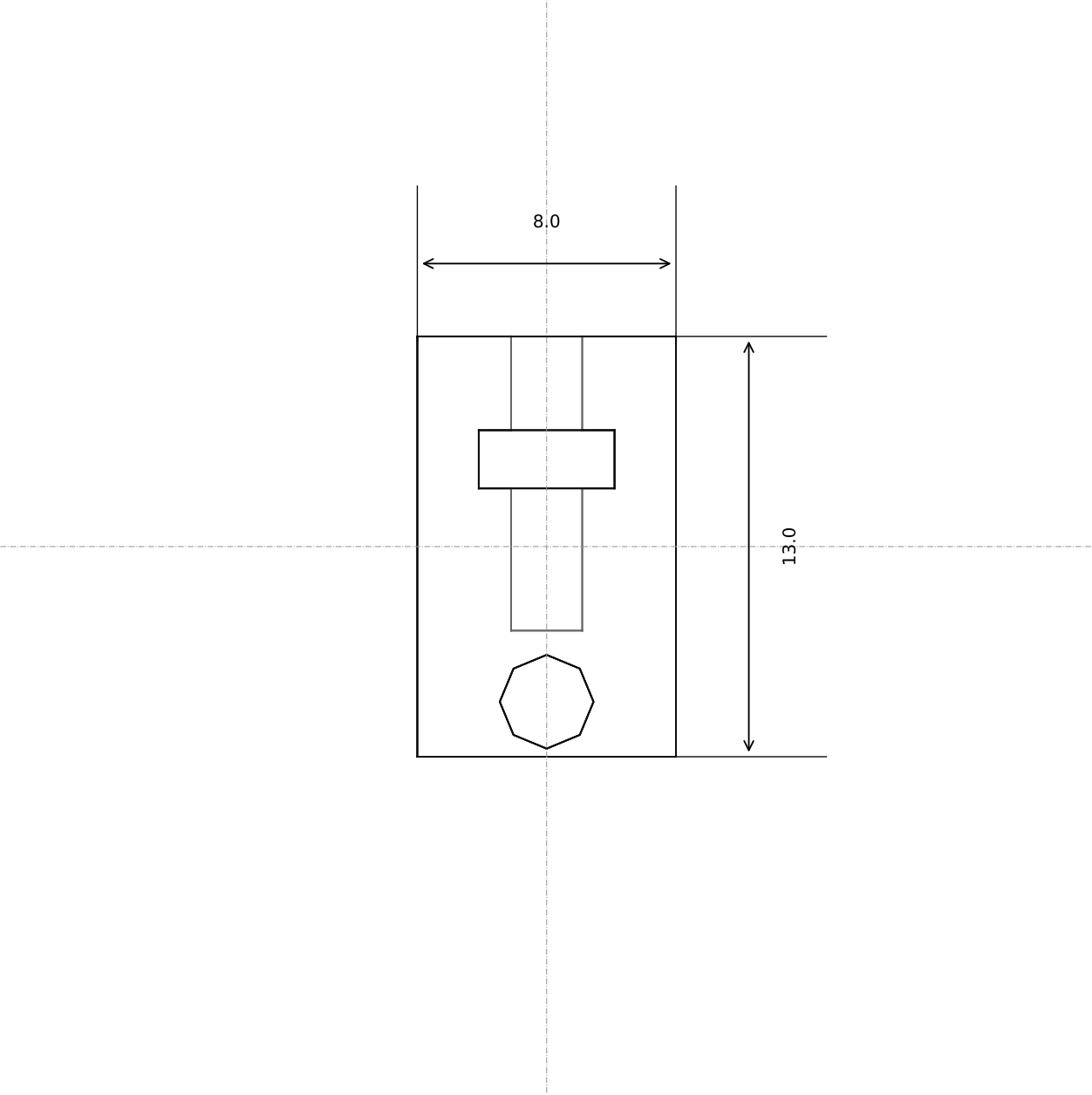
Halo liner (white)

Cadrane · sheet 4 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

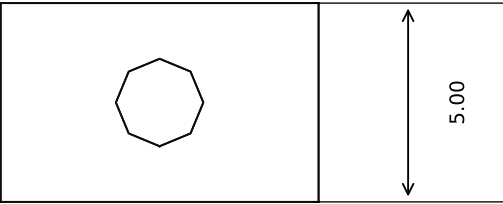
Material: white PETG
Process: v16: print flat on its floor, 0.16 mm; no supports
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: halo_liner
Overall 181.0 x 181.0 x 3.4
Volume 1.85 cm³
Scale: see each view (fit)

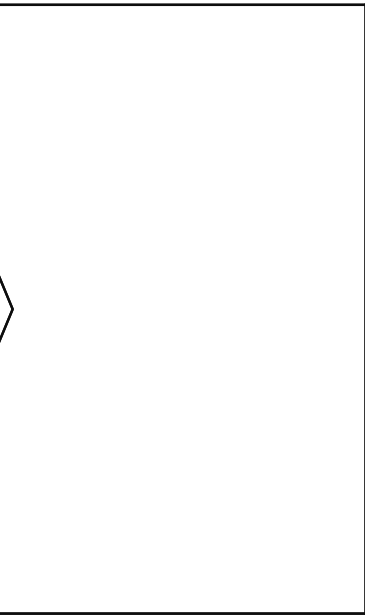
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



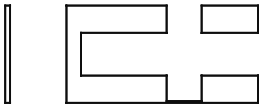
SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 2 to 6, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 13.0 (radial) x 8.0 x 5.0 on the plate top at 270°, r 71.0-84.0 nominal; Ø2.9 press hole for the Ø3 m6 x 16 axle pin 1.7 from its outer end (the pin to z 24.2)
- Ø2.2 hole from the inner face at z 10.5 for the M2 x 18 release screw; an M2 nut in a 4.2 slot from the top, 3.0-4.6 behind the face
- spring Ø4.0 x 16.0 free, 0.8 N/mm on the screw between the channel's end wall and the block: 4.0 N seated, 7.2 N parked
- 0.2 to the channel's walls and roof; slides on the anodised plate

the 60 — v16

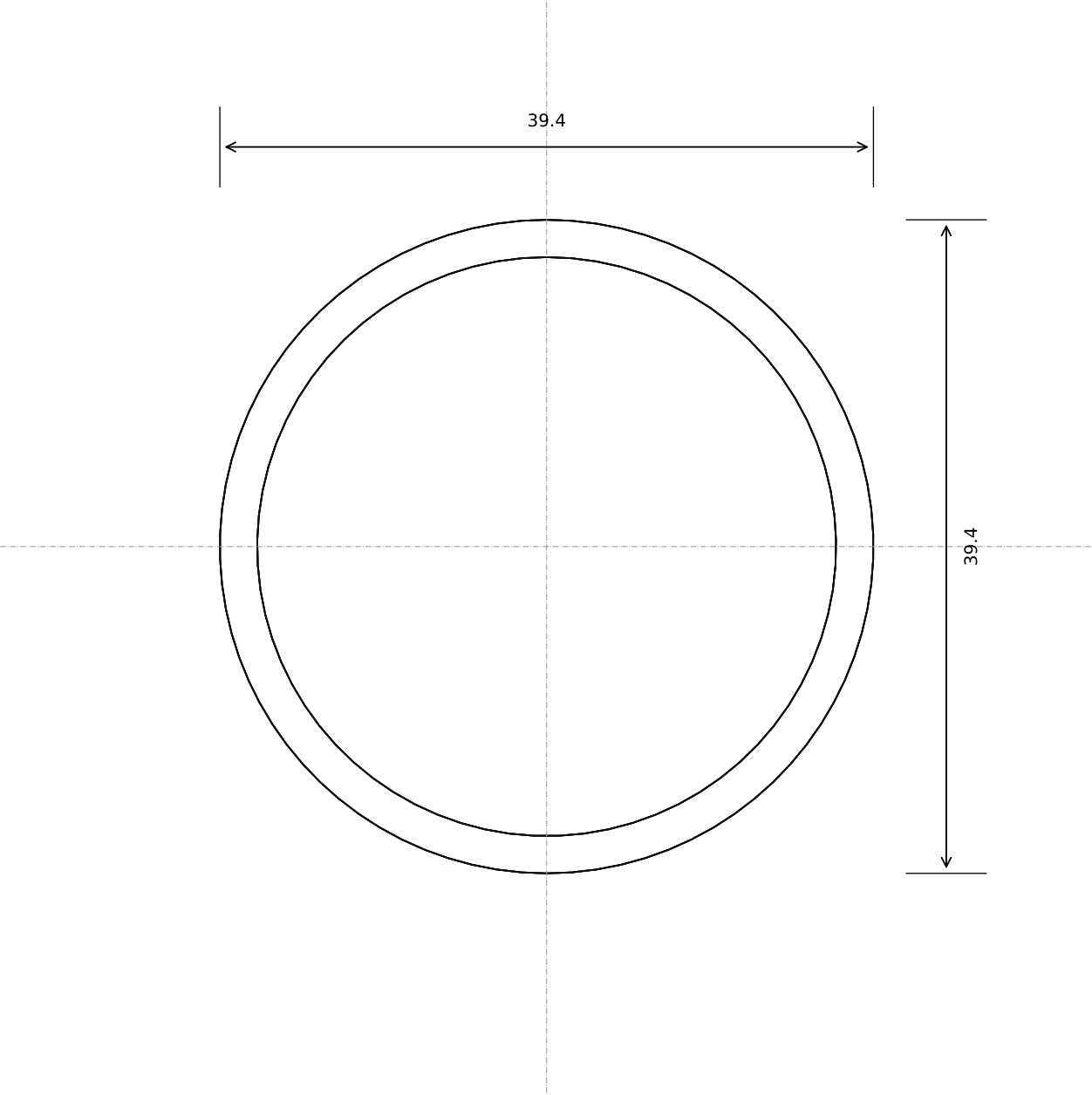
Sprung wheel block

Cadrane · sheet 5 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

Material: PETG, 4 perimeters
Process: v16: print flat on its bottom face, 0.12 mm
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: wheel_block
Overall 8.0 x 13.0 x 5.0
Volume 0.42 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 10 to 22, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. ID 35.0 -0.05 (press) x OD 39.4 x 3.5, on the bell's top (z 14.7-18.2); 0.3 lead-in chamfer
- 2. the 0.6 silicone band on it at z 14.90-17.8 reaches the bore (r 88.20); the bell itself stops at r 86.0, inside the wall's outer face and 0.5 inside the halo liner
- 3. bore/collar ratio 4.48

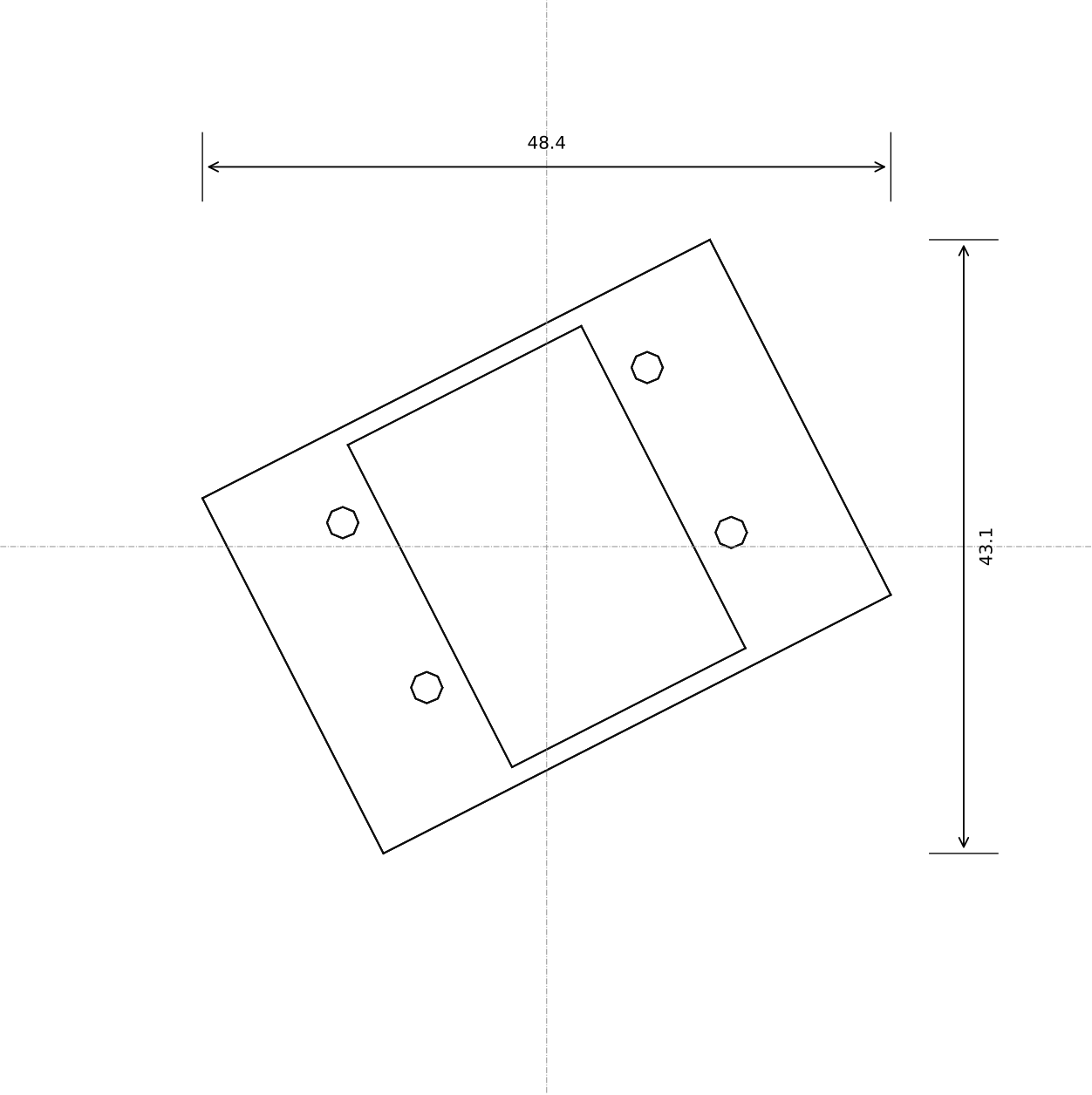
the 60 — v16 Drive collar (on the bell)

Cadrane · sheet 6 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

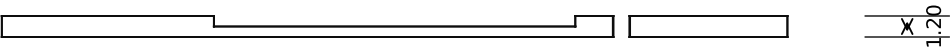
Material: PETG (prototype) / POM (production)
Process: v16: print axis vertical, 0.12 mm
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: drive_collar
Overall 39.4 x 39.4 x 3.5
Volume 0.91 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 12 to 26, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 40.0 x 28.0 x 1.2 at z 16.6-17.8, centred 0.0 toward the ports of the audio board's centre; four Ø2.2 on the audio's stud pattern (24 x 13); on 4.4 tube spacers above the audio's nuts
- a 0.6 pocket: the ESP32-S3-Zero class MCU (25.0 x 18.0) between the four stud nuts, USB-C toward the west end (the level shifter is on the plate at az 322.0, r 69.5)
- no screw touches the MCU: the keeper's rails hold it by its edges

the 60 — v16

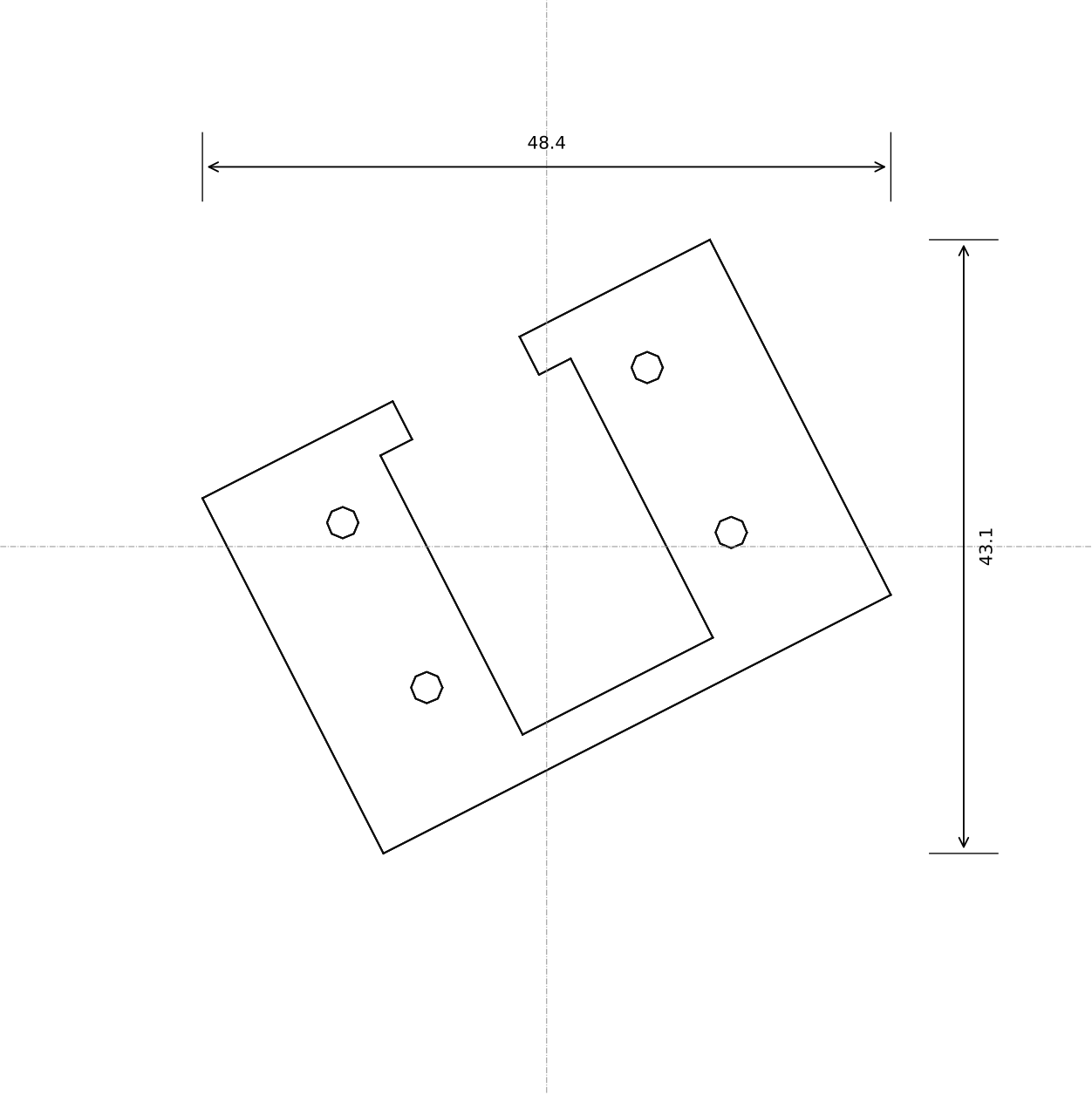
MCU tray (over the audio board)

Cadrane · sheet 7 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

Material: PETG
Process: v16: print flat on its underside, 0.16 mm
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: mcu_tray
Overall 48.4 x 43.1 x 1.2
Volume 1.05 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 12 to 26, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

1. 40.0 x 28.0 x 1.2 on the tray (z 17.8-19.0), clamped by the audio studs' top nuts; a window over the MCU's parts (1.5 rails); its USB-C end open

the 60 — v16

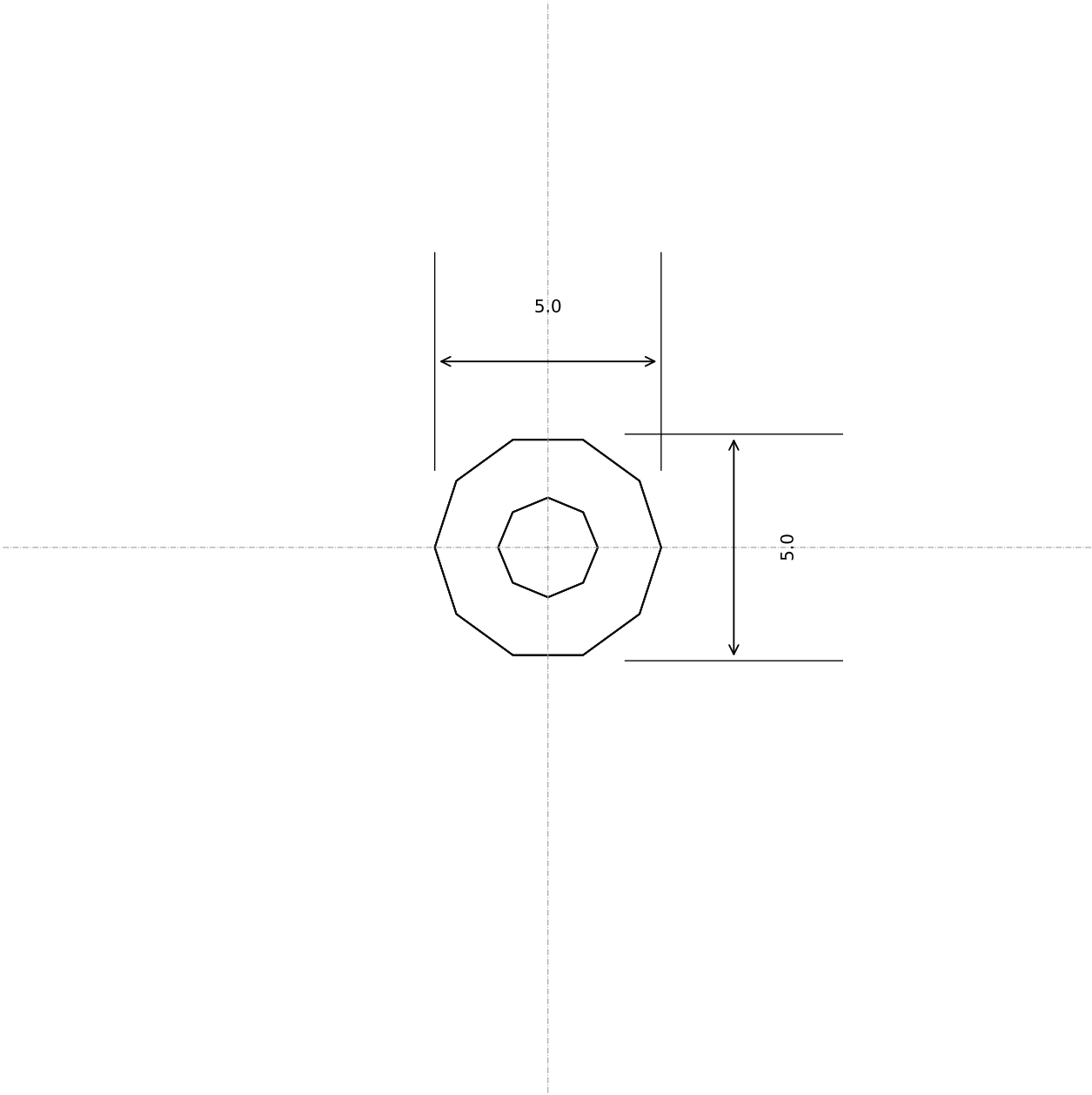
Tray keeper

Cadrane · sheet 8 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

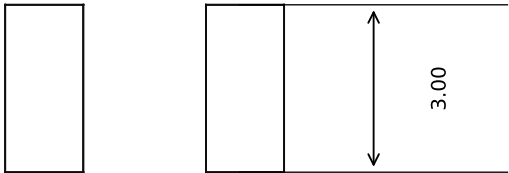
Material: PETG
Process: v16: print flat, 0.16 mm
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: tray_keeper
Overall 48.4 x 43.1 x 1.2
Volume 0.89 cm³
Scale: see each view (fit)

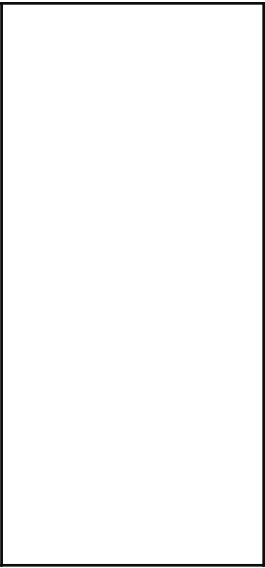
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



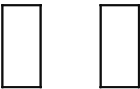
SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 1 to 4, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. Ø5.0 tube, bore 2.2: 3.0 under the driver and haptic breakouts, 1.0 under the audio board, 4.4 between the audio's nuts and the tray
- 2. the countersunk M2 stud passes through; the nut above the board clamps the stack

the 60 — v16

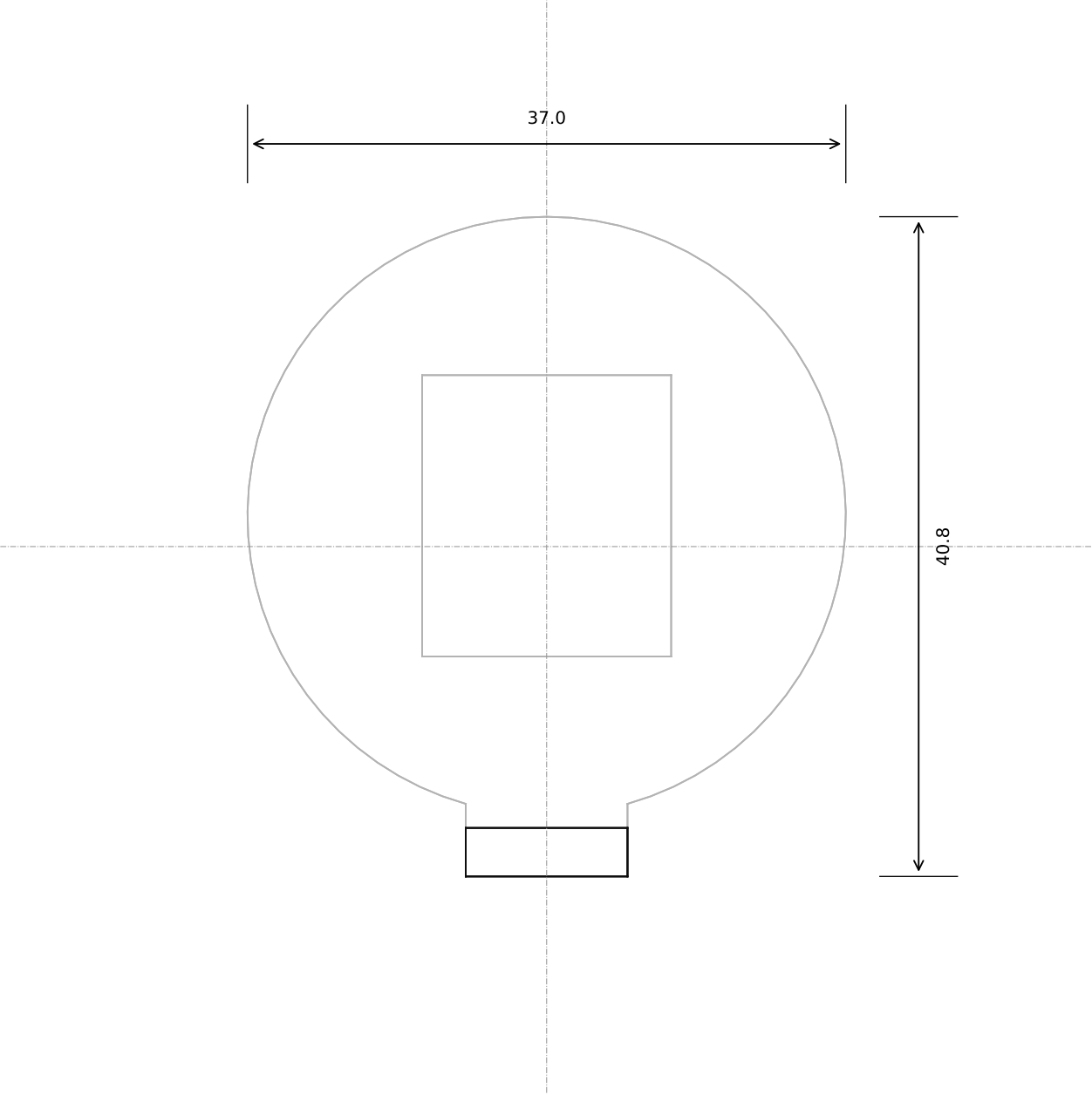
Board spacer (family)

Cadrane · sheet 9 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

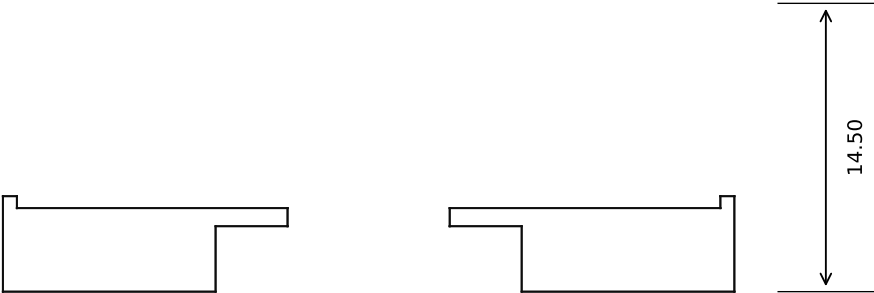
Material: PETG
Process: v16: print axis vertical, 0.12 mm; 3.0 x 4, 1.0 x 4, 4.4 x 4 mm. Tolerance ±0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: board_spacer
Overall 5.0 x 5.0 x 3.0
Volume 0.05 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)

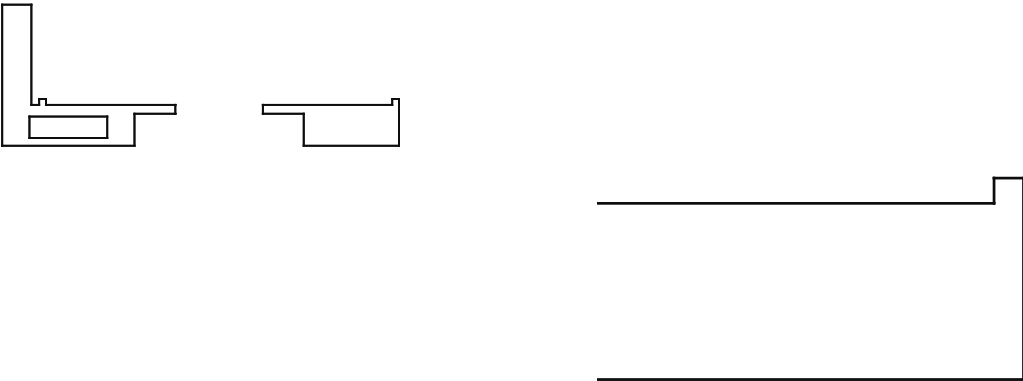


SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 9 to 20, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. shoe $\varnothing 37 \times 4.2$; locating rim $\varnothing 35.6/37 \times 0.6$
- 2. pocket $17.4 \times 15.4 \times 3.3$ for the MT6701 module (v16); sight hole $\varnothing 9$; a 2.4×2.2 notch through the shoe's inboard wall for the sensor lead
- 3. push tab 3.0×10.0 to $z\ 14.5$ (v11: 28.5 through the deck; the servo is now beside it on the plate)
- 4. travel 2.4 radial in the plate's stadium hole

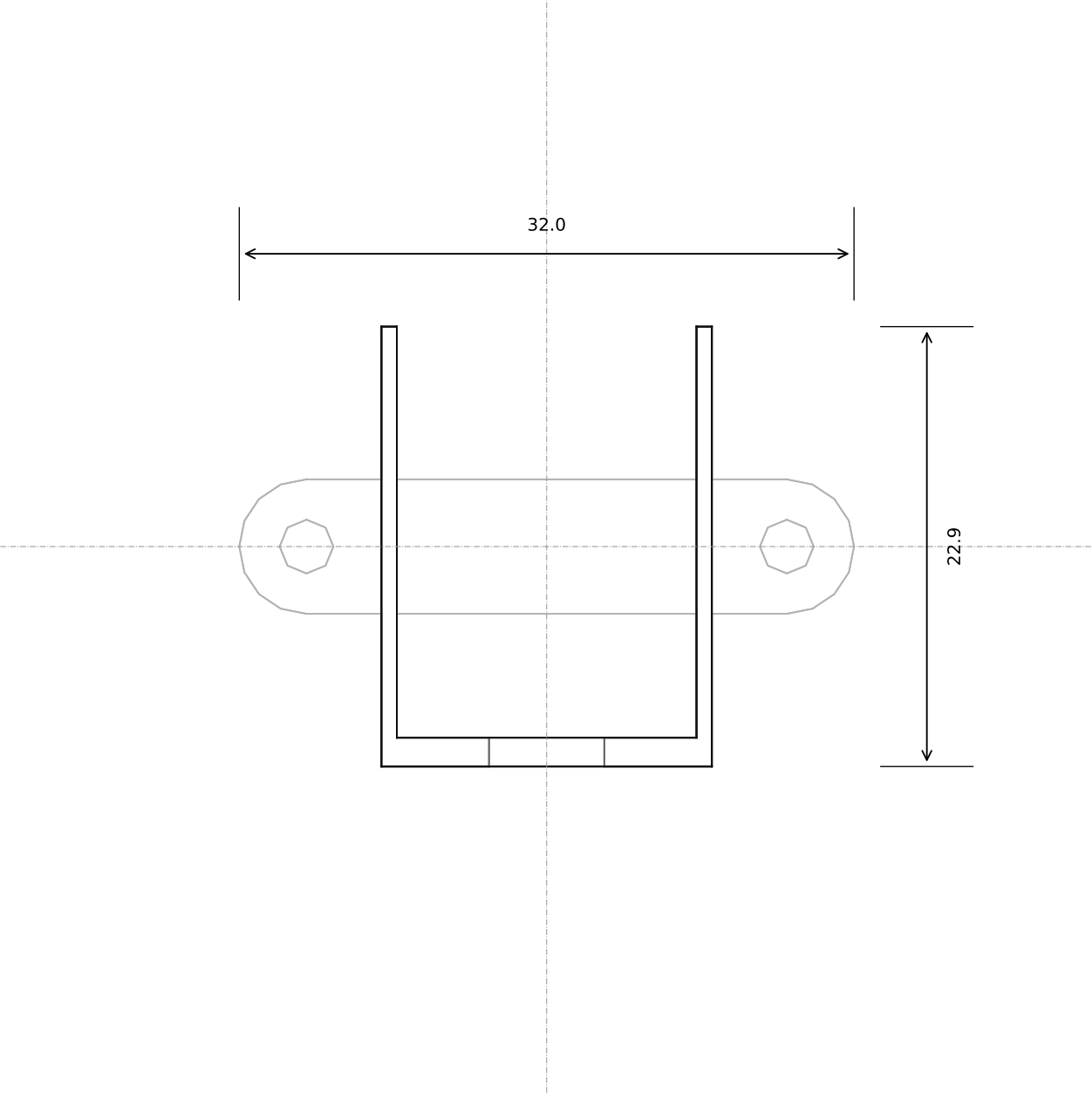
the 60 — v16

Motor carriage

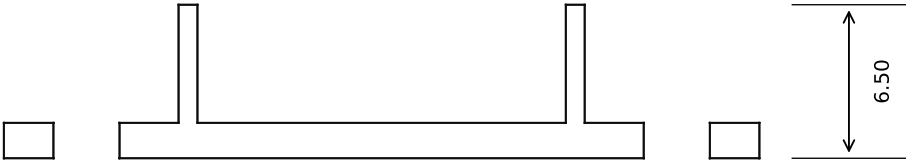
Cadrane · sheet 10 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

Material: PETG	Part: carriage
Process: print standing on the shoe, 0.16 mm; no supports (12 mm bridge over the 37.0 x 40.8)	Overall 37.0 x 40.8 x 14.5 perimeters for the tab
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	Volume 4.06 cm ³
	Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)

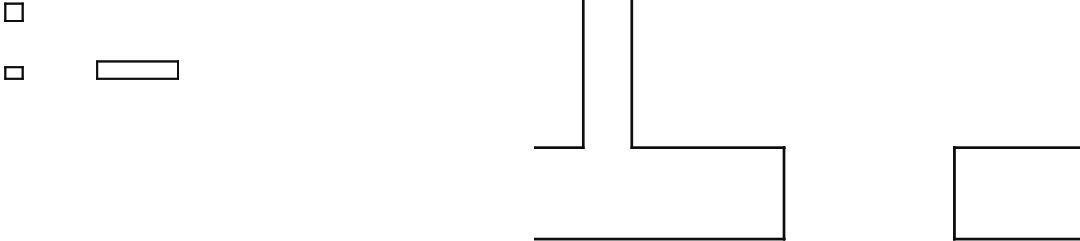


SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 8 to 18, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. floorless frame round the AGFRC envelope 21.4 x 15.2 x 6.0; walls 1.0, 6.5 tall; the servo lies on the plate at r 21.8-43.2, az 90
- 2. open toward the carriage tab; lead notch in the inner end wall
- 3. 2 ears at t ±12.5, M2.5 x 4 into tapped blind holes in the plate's web

the 60 — v16

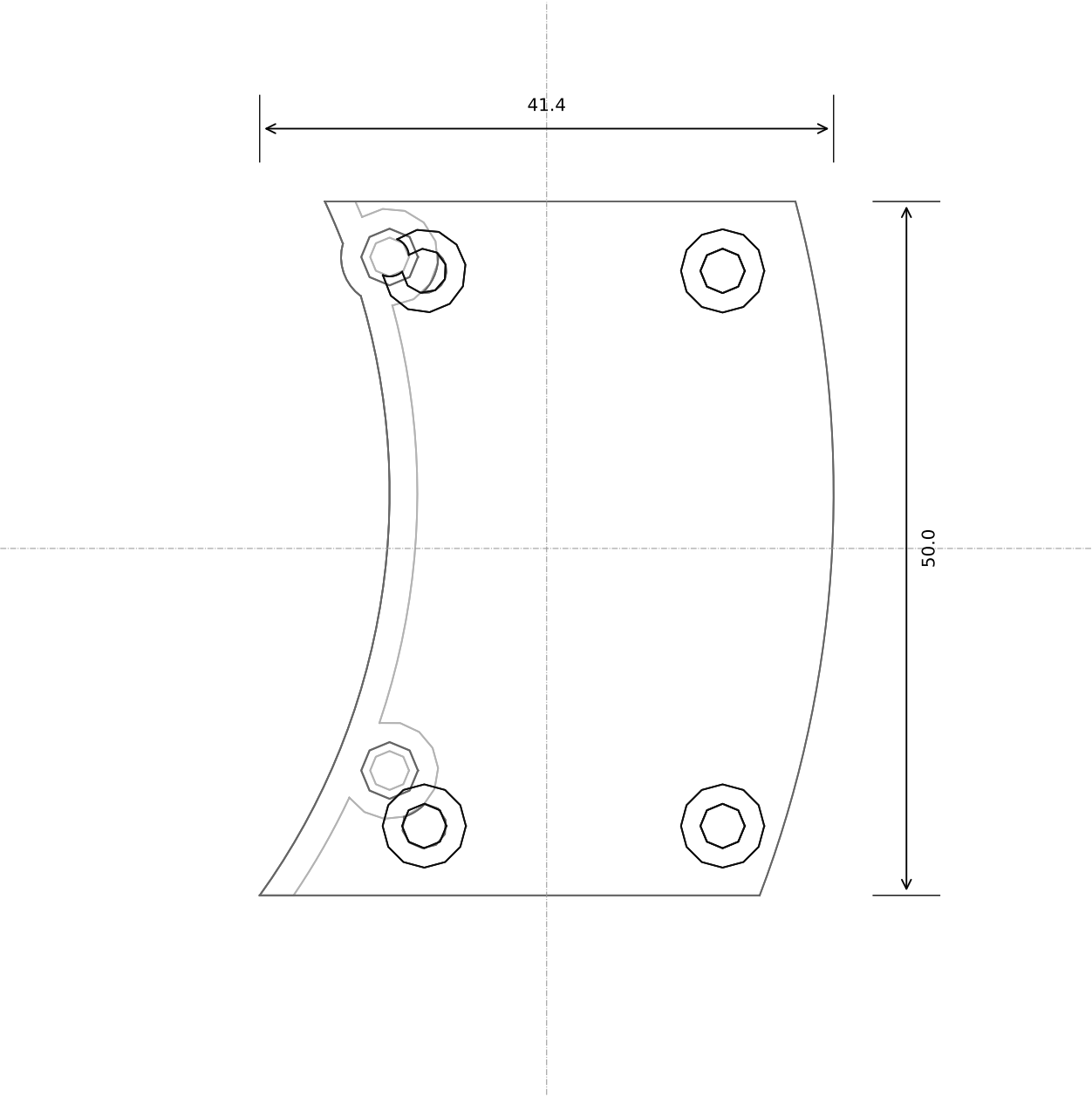
Servo frame (on the plate)

Cadrane · sheet 11 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

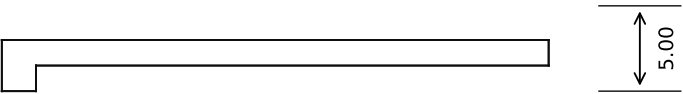
Material: PETG
Process: print flat, 0.20 mm; no supports
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: servo_mount
Overall 32.0 x 22.9 x 6.5
Volume 0.64 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 10 to 23, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. slab 1.5 thick at z 9.5-11.0, r 50-82, t -29..21, bridging the port slot; its outer end rests on the port face's rail (z 9.5)
- 2. foot strip r 50-52 on the plate; two csk ears at [(49.6, 17.0), (49.6, -20.0)], M2.5 into the web
- 3. four Ø6 bosses 2.0 tall with M2 inserts for the connect board (30.0 x 46.0, holes 3 in from its inner corners, 5.5 from its outer)
- 4. the -y inner corner cut back to x 45.5 (the audio board's USB plug is there)

the 60 — v16

Connect-board bracket

Cadrane · sheet 12 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

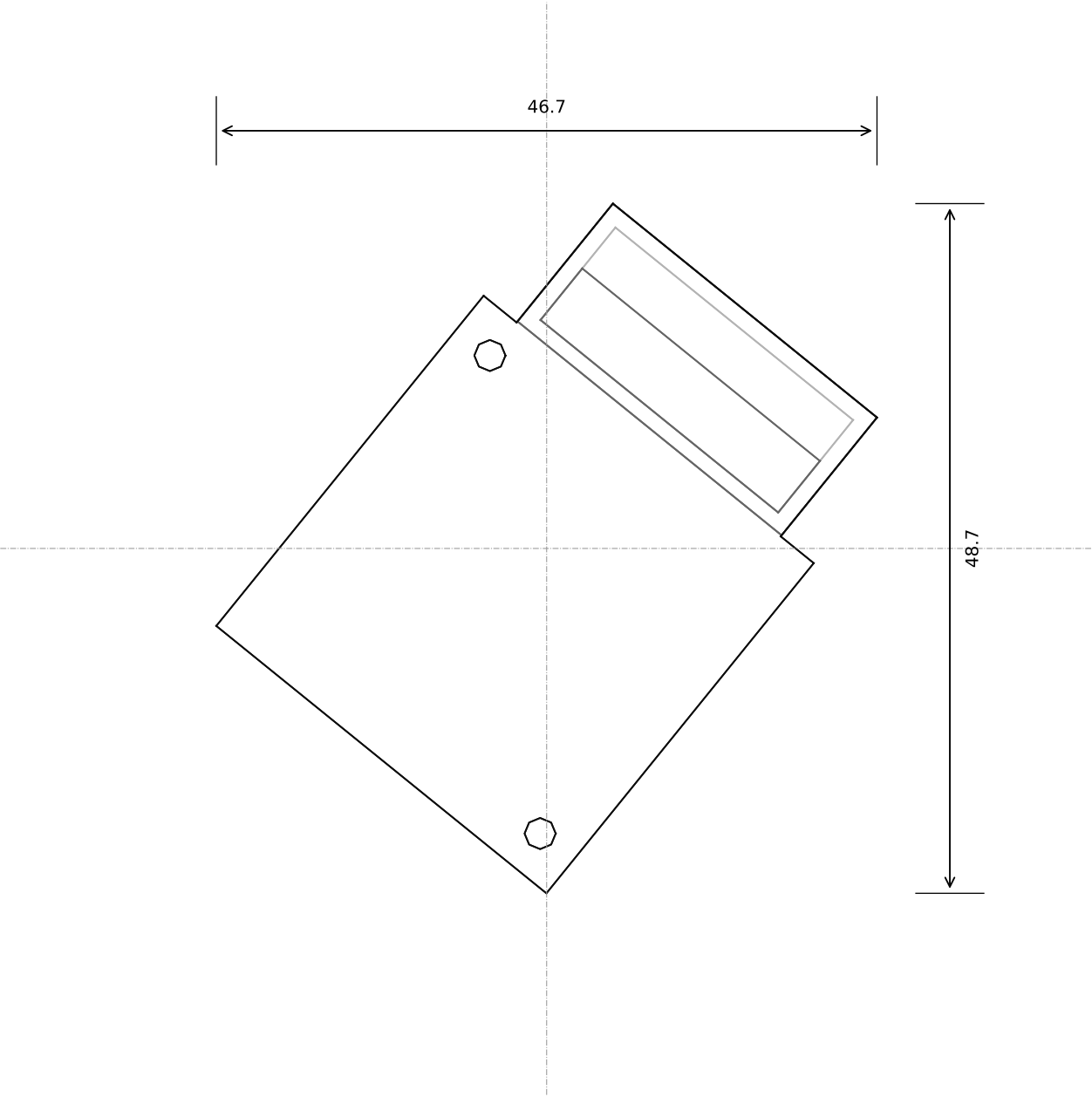
Material: PETG

Process: print standing on its straight -y edge (slab vertical, feet and bosses clear by 41.40 x 20.00 x 5.00 supports
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: connect_bracket

Size: 41.40 x 20.00 x 5.00
Volume 2.80 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 12 to 25, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. plate 30 x 30 x 1.2 on the blower's top (z 18.3-19.5) - v16: a plain clamp plate, no board on it; two Ø2.2 through-holes on the blower's 24 x 24 diagonal: M2 x 16 pan heads through the plate and the blower into the plate's piers
- 2. hood neck: 24 wide, r 66-76.8, walls 1.2, open at the bottom over the trench and toward the blower's outlet (21 x 7); on a 0.3 foam gasket on the plate top
- 3. blower at az 51, r 51 (Delta BFB0305HA-C 30 x 30 x 10, inlet down over the Ø22 hole in the web)

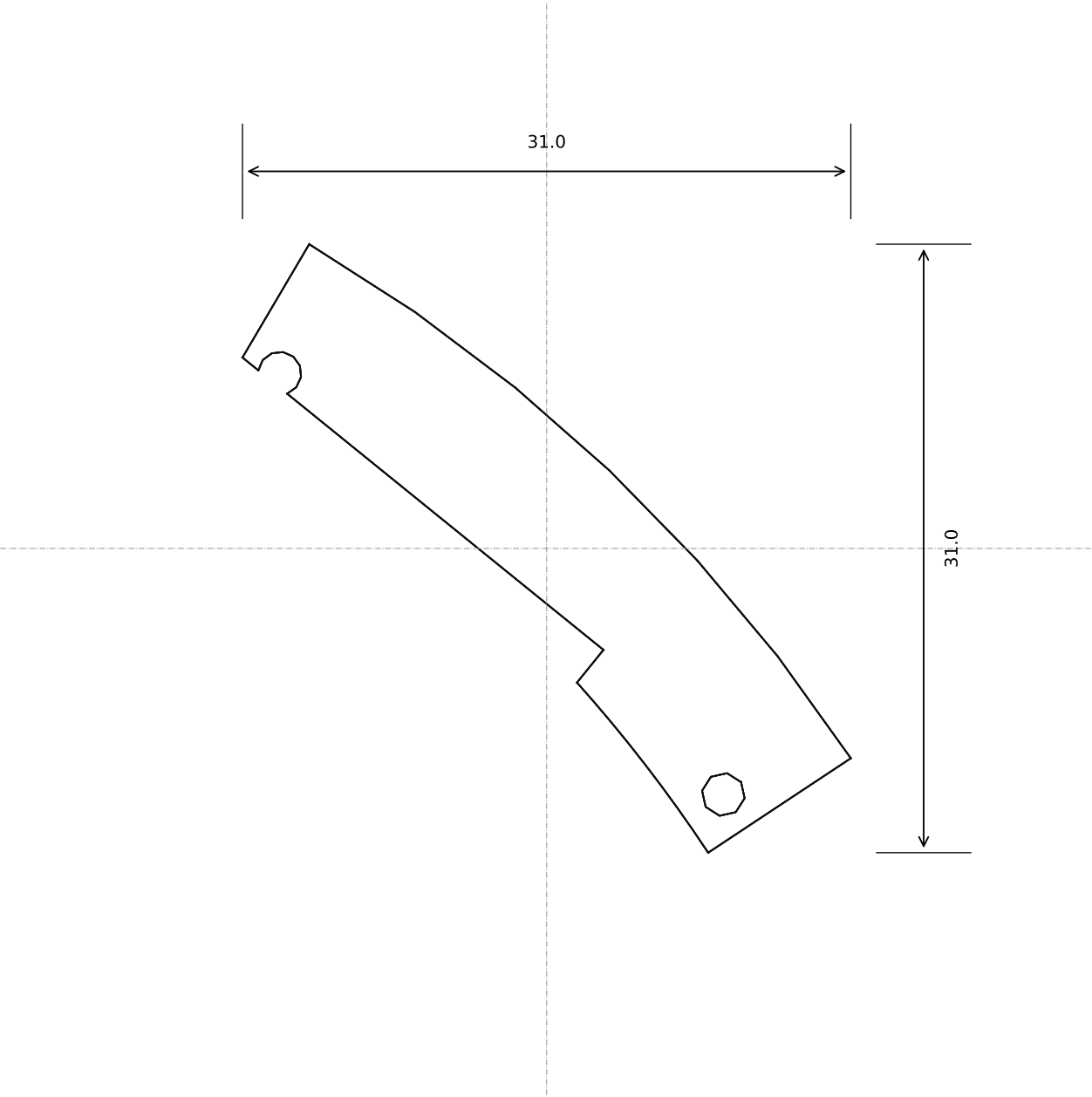
the 60 — v16

Blower saddle + hood neck

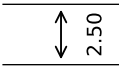
Material: PETG
Process: print on its flat top face, 0.16 mm; no supports
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: **blower_saddle**
Overall 46.7 x 48.7 x 11.2
Volume 2.76 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 8 to 17, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):
1. 2.5 thick at z 8.3-10.8, az 33.5-59.5, r 75.8-84.5; butts the saddle's neck (0.15 clear); two Ø2.2 at az [35.0, 58.0], r 78.1: M2 x 5 into the ring's top; on the same 0.3 gasket

the 60 — v16

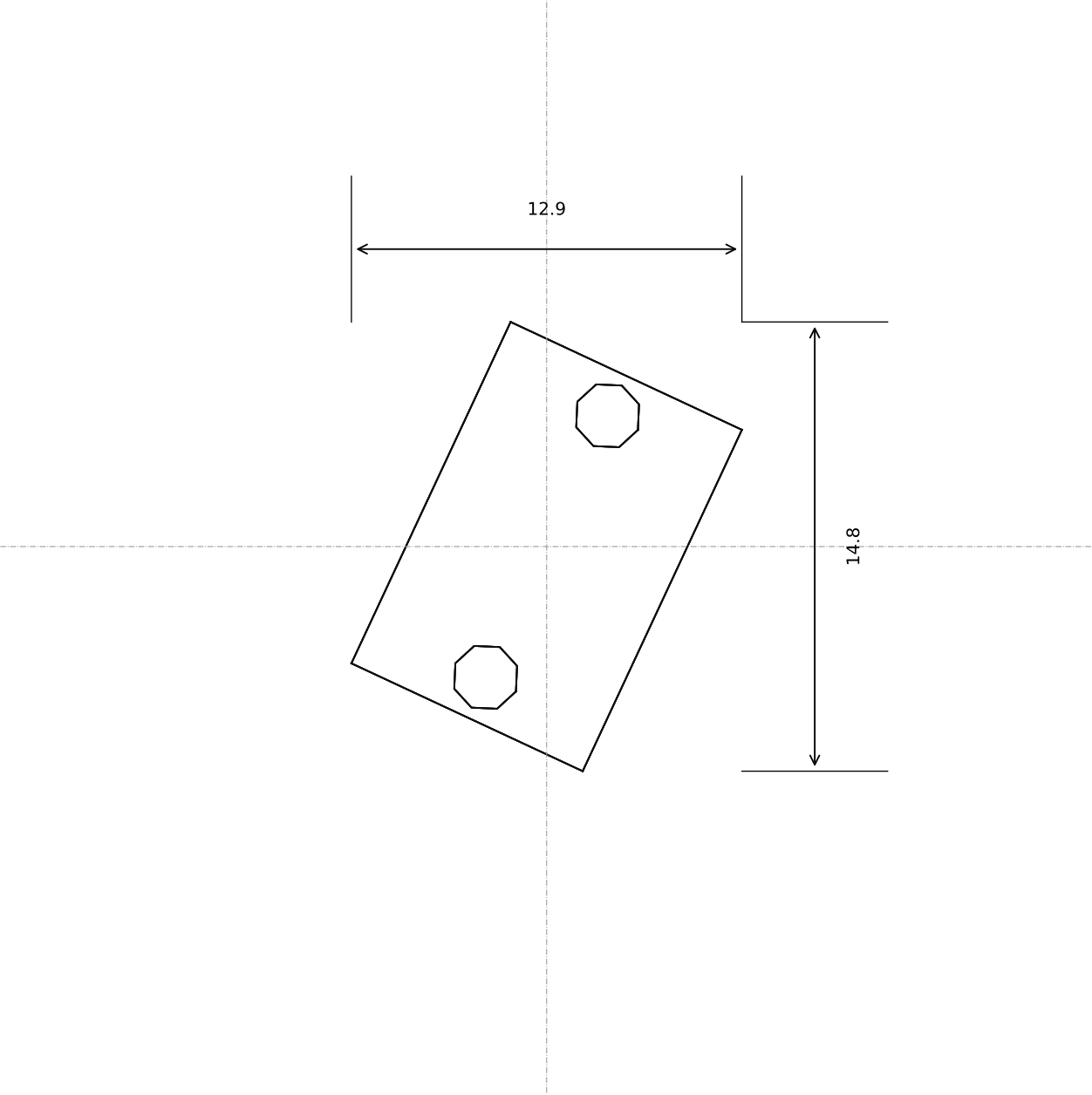
Hood lid (over the trench foot)

Cadrane · sheet 14 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

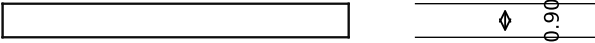
Material: PETG
Process: v15: print flat, 0.16 mm; no supports
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: hood_lid
Overall 31.0 x 31.0 x 2.5
Volume 0.68 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 3 to 8, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):
1. 8.4 x 12.4 x 0.9 (nominal; family [0.7, 0.8, 0.9, 1.0, 1.1]), two Ø2.2 at ±4.75; sits on the seat flange under the encoder board, sets the 2.0 optical gap

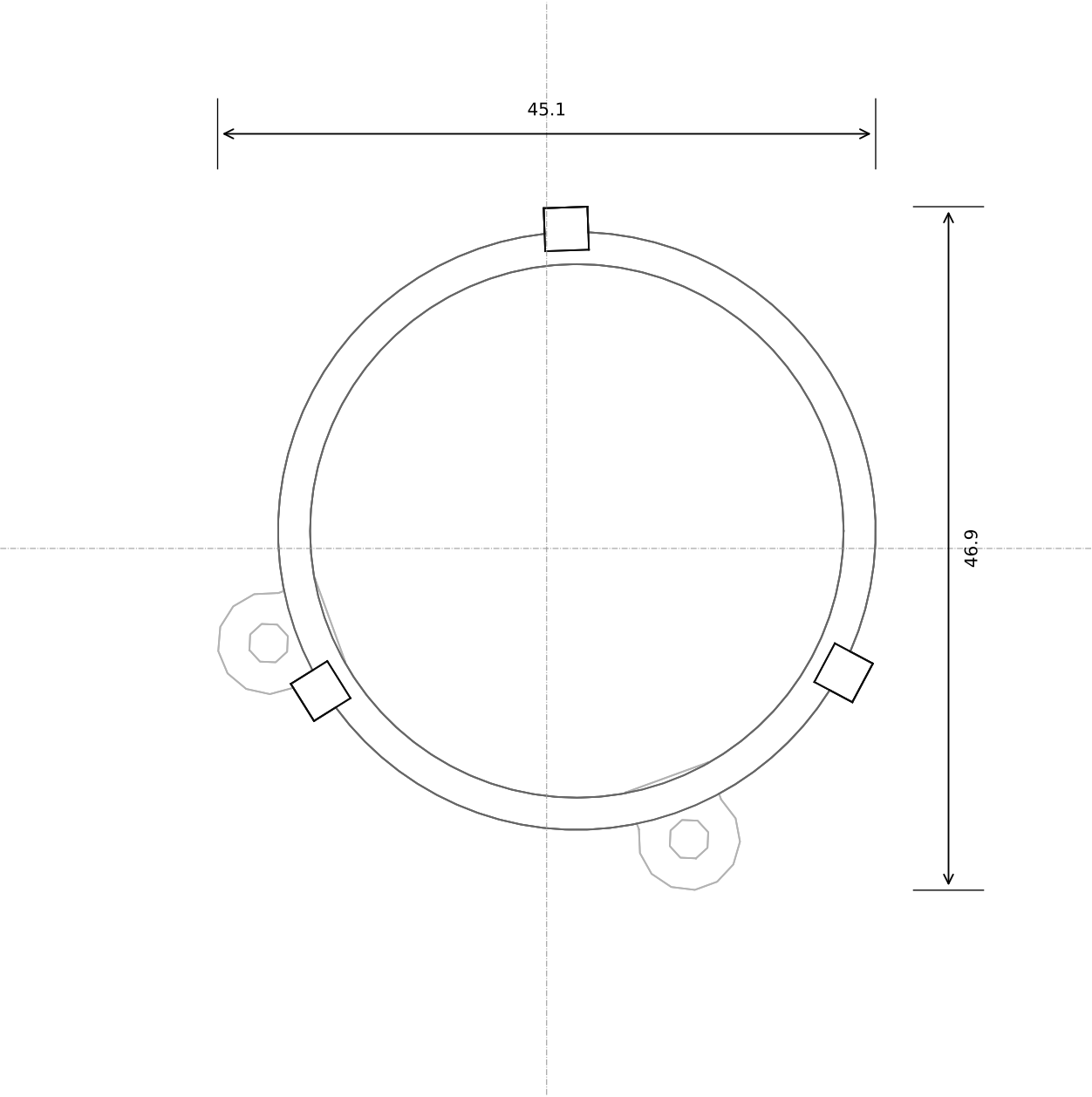
the 60 — v16 **Encoder shim (family)**

Cadrane · sheet 15 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

Material: PETG
Process: v15: print flat, 0.10 mm; one of each height in ENC_SHIM_FAMILY
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: encoder_shim
Overall 12.9 x 14.8 x 0.9
Volume 0.09 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 11 to 25, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. ring r 18.3-20.5 x 4.45 under the Ø40 flange
- 2. three snap fingers at [92.0, 212.0, 332.0]° from the speaker's centre (device azimuths)
- 3. two ears at r 22.5, [200.0, 290.0]° from the centre; M2.5 x 4 from above into the plate's web (v11: M3 from below)
- 4. speaker centre (-36, 47) - az 127°, r 59.2

the 60 — v16

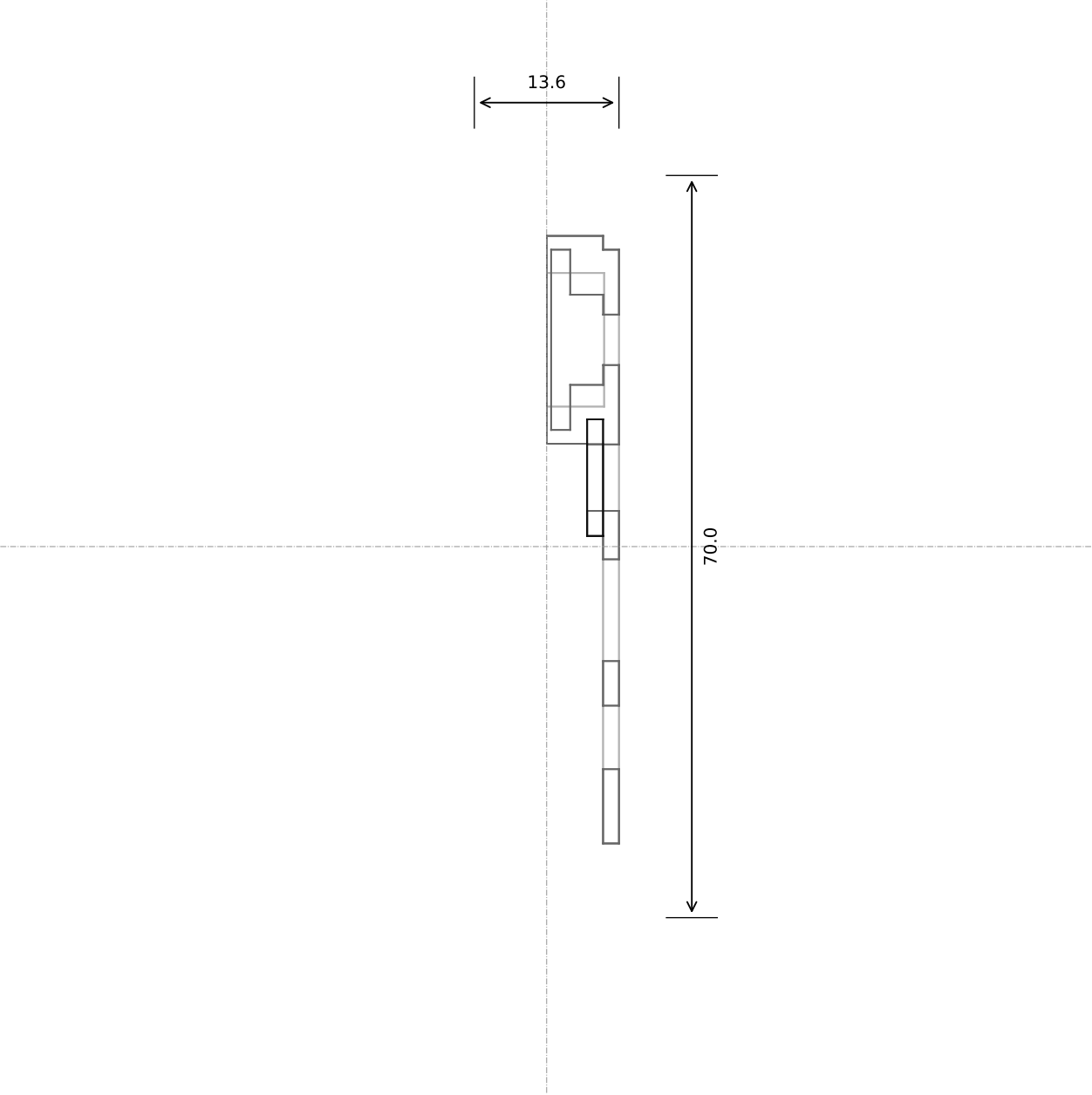
Speaker cradle

Cadrane · sheet 16 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

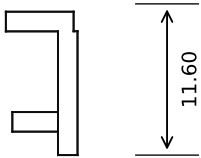
Material: PETG
Process: print flat, 0.20 mm; no supports (fingers full height)
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: speaker_cradle
Overall 45.1 x 46.9 x 7.2
Volume 1.44 cm³
Scale: see each view (fit)

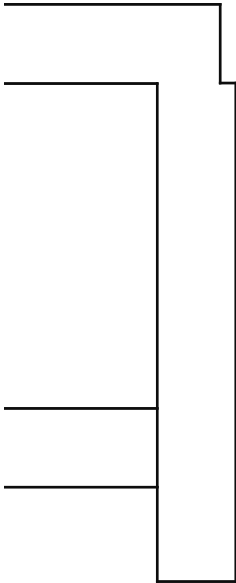
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 3 to 9, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. 56.0 wide x 1.5 thick at r 83.30, z -1.5-8.0
- 2. barrel nose Ø6.2 at t -18.0, z 3.5; USB-C opening at t -6.0, z 4.5; jack Ø6.3 at t 6.5, z 4.6; light aperture Ø4.8 at t 19.5
- 3. shelf for the USB-C board; a 3.0 slot holder for the Adafruit 4162 light-sensor breakout (16.6 x 16.5, standing from z 1.0 to 17.5, sensor at the hole) with a pocket for its sensor package; a 3.0 boss round the panel jack's Ø6.3 hole; rail along the top on the plate (1.5 tall, 70 wide, inner edge r 79.3, flat front in the faces plane, Material: PETG

the 60 — v16

Rear port face

Cadrane · sheet 17 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

Process: print lying on its outer face, 0.16 mm; no supports (the socket holder is printed on the other side of the plate)
mm. Tolerance ±0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

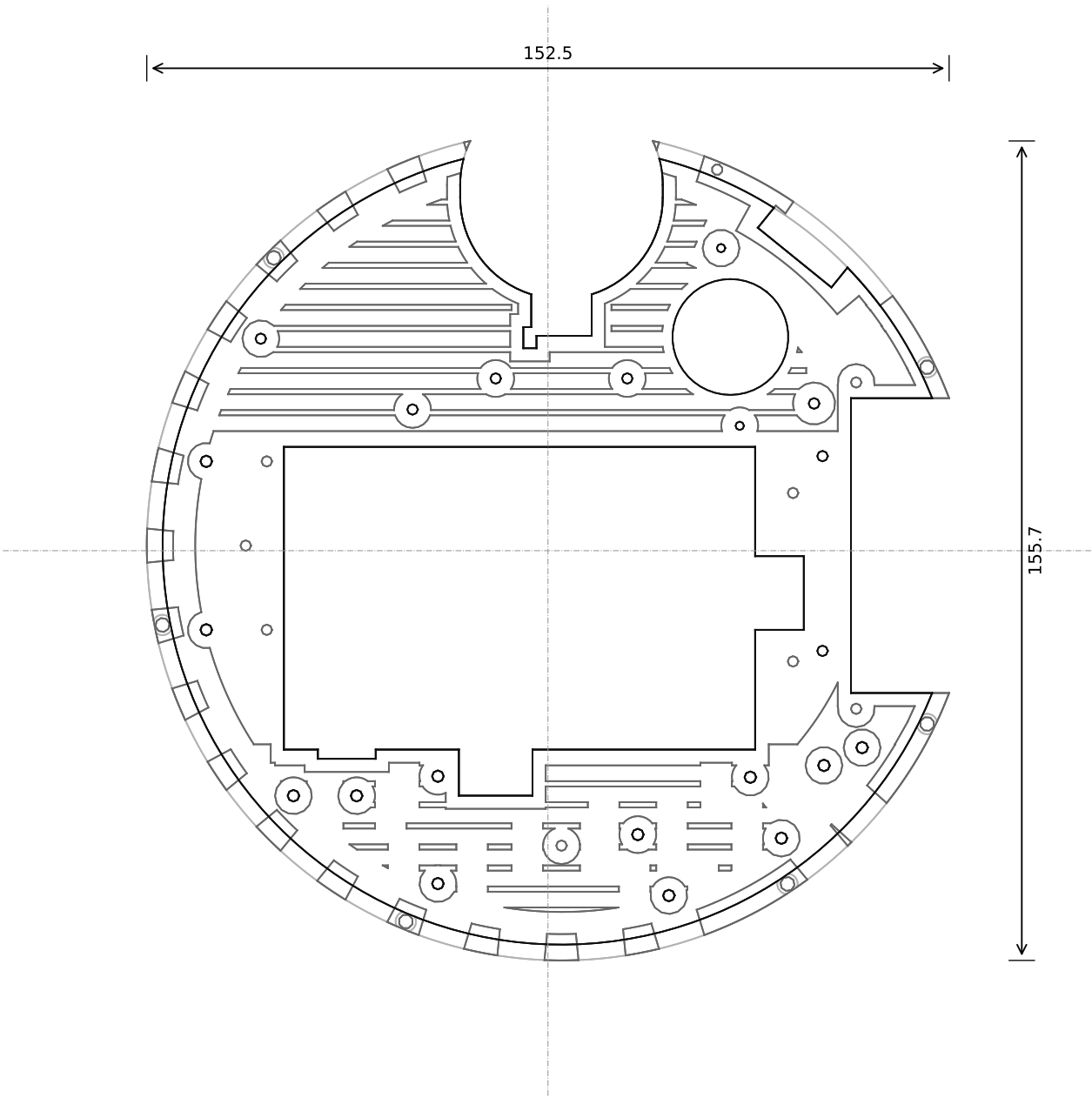
Part: port_face

Overall size: 70.0 x 11.6

Volume 1.65 cm³

Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 46 to 78, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. $r < 78.8 \times 8.0$: 2 web (z 6-8), 5 duct (z 1-6) closed by a 1 mm plate in a recess on a 0.2 gasket in a groove; $\varnothing 7$ piers 2.5 into the duct under every screw from the top; rebate r 75.8-78.8, z 5-8 for the ring's flange on a 5 shoulder, 7 x M2.5 csk from below at r 77.30; the ring's bond screw M2.5 tapped 3.0 into the shoulder at r 77.30
- 2. duct: collector r 70-73.8 (front and sides), 21 straight channels 3 x 5 on a 4 pitch beside the Pi window, rear plenums r 59+; 2.5 walls round every through-cut; intake passages as closed tunnels z 1-4.5 through the duct wall and shoulder (18 x 9 wide) + 2 exhaust tunnels 10 wide at az [310.0, 318.0]
- 3. BLOWER (v15, fitted): $\varnothing 22$ inlet hole through the web at az 51, r 51; discharge trench 18 wide from r 70.4 out through the duct wall and the shoulder, widening to az 37-56 (open-topped, the hood seals it); the 2 mm wall at r 68.4-70.4 keeps it off the suction side
- 4. Pi window (Pi at (-52, -38), with the USB-C receptacle, HDMI plug and USB-A plug notches - v16: no plug-head notch, the Pi is powered through its header), carriage hole, port slot 56.0 wide; 6 x M2.5 tapped blind 3.5 deep from the top (servo frame, connect bracket, speaker cradle) + 2 x M2 (the blower) + THE chassis bond screw
- 5. v16: flat top face - no ribs (the two v15 ribs are deleted)

the 60 — v16

Base plate - aluminium core (the heatsink)

Material: 6082 aluminium, CNC two-sided; EXTERNAL faces black hard anodized	Part: base plate
Process: machined, not printed (print a PETG stand-in without the duct for the fit)	Overall 152.5 x 155.7 x 8.0
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	Volume 45.73 cm ³
	Scale: see each view (fit)

Technical drawing of a circular architectural element, likely a dome or vaulted ceiling section. The drawing shows a cross-section of the structure, which is semi-circular with a flat base. The outer edge is defined by a series of small, rectangular blocks, possibly representing masonry or tiles. The inner surface is smooth and curved. A horizontal dashed line passes through the center of the dome, and a vertical dashed line passes through the base. Two dimensions are indicated: a horizontal dimension of 185.2 at the top, and a vertical dimension of 189.4 on the right side. The drawing is a black and white line drawing with no shading.



8.00

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[

1. r 78.8-92.7 x 8.0; 90 obround openings 2.0 x 4.5 (z 1.55-6.05) on 3° through the 3.7 outer wall, az 1.5 + 3k, mirrored above 0-180, none across the port face (342-18): 64 intake (60-300) + 26 exhaust (18-60, 300-342); 0.3 x 45° polished at every mouth; the upper land plain (no reeding - Ryan, 7 Sep); 0.3 chamfers on both of its edges; 0.35 above the openings' chamfered mouths	Cadrane - sheet 19 of 20 - 2026-09-08 - drawn from src/params.py + model.py (build123d) the outer wall (r 89.00-92.7) opens into it all round both arcs; the blower's trench: flange notched and inner land cut away az 37-56 to r 83.00, the groove is 0.3 deep Material: stainless steel 304/316, machined; BARE: electropolish (the 0.3 chamfers are electropolished) Process: machined, not printed	Part: r 185.2 x 189.4 x 8.0 Volume 32.24 cm³
2. underside groove behind the openings: r 84.50-89.00 (intake arc) / r 82.50-89.00 (exhaust arcs), z 0-6.55, walls 1.2° at 60 and 300; the pad closes it; the 0.8 undercut under the openings' pad is 0.3 deep		
3. inward flange r 75.8-78.8, z 5-8, 7 x M2.5 tapped blind from below at r 77.30, az [26.0, 78.75, 135, 191.25, 247.5, 303.75, 334.0]; the dedicated BOND screw's Ø2.8 clearance is 0.3		
4. passages across the inner land (r 78.8 to the groove), 3.5 deep, the pad closes them: 18 intake 9 wide (567 mm²) on the 11.25° grid, 2 exhaust 10 wide at az [310.0, 318.0] (the -y circuit; the +y circuit exhausts through the trench)		
5. 3 x M2.5 csk from below at r 80.8, az [160, 240, 328] (the structure's pillars); 2 x Ø1.6 tapped M2 at r 81.8, t ±32 (the port face's rail); port slot 56.0 wide		
6. v16 (HALO-OPTICS): the opal ring's seat - a 0.5 rebate in the top from r 90.50 outward and a 0.8 shelf r 92.7-94.7 at z 6.7-7.5 standing 2.0 proud (0.2 polished chamfers on its edges)		

Rim ring (stainless)

Intake (60-300) + 26 exhaust (18-60, 300-342); 0.3 x 45° polished at every mouth; the upper land plain (no reeding - Ryan, 7 Sep); 0.3 chamfers on both
Cadrane · sheet 19 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

Material: stainless steel 304/316, machined; BARE: electropolish (the 0.3 micron max avg. **STEEL** mouth and both edges), then brush the

Process: machined, not printed

Overall 185.2 x 189.4 x 8.0

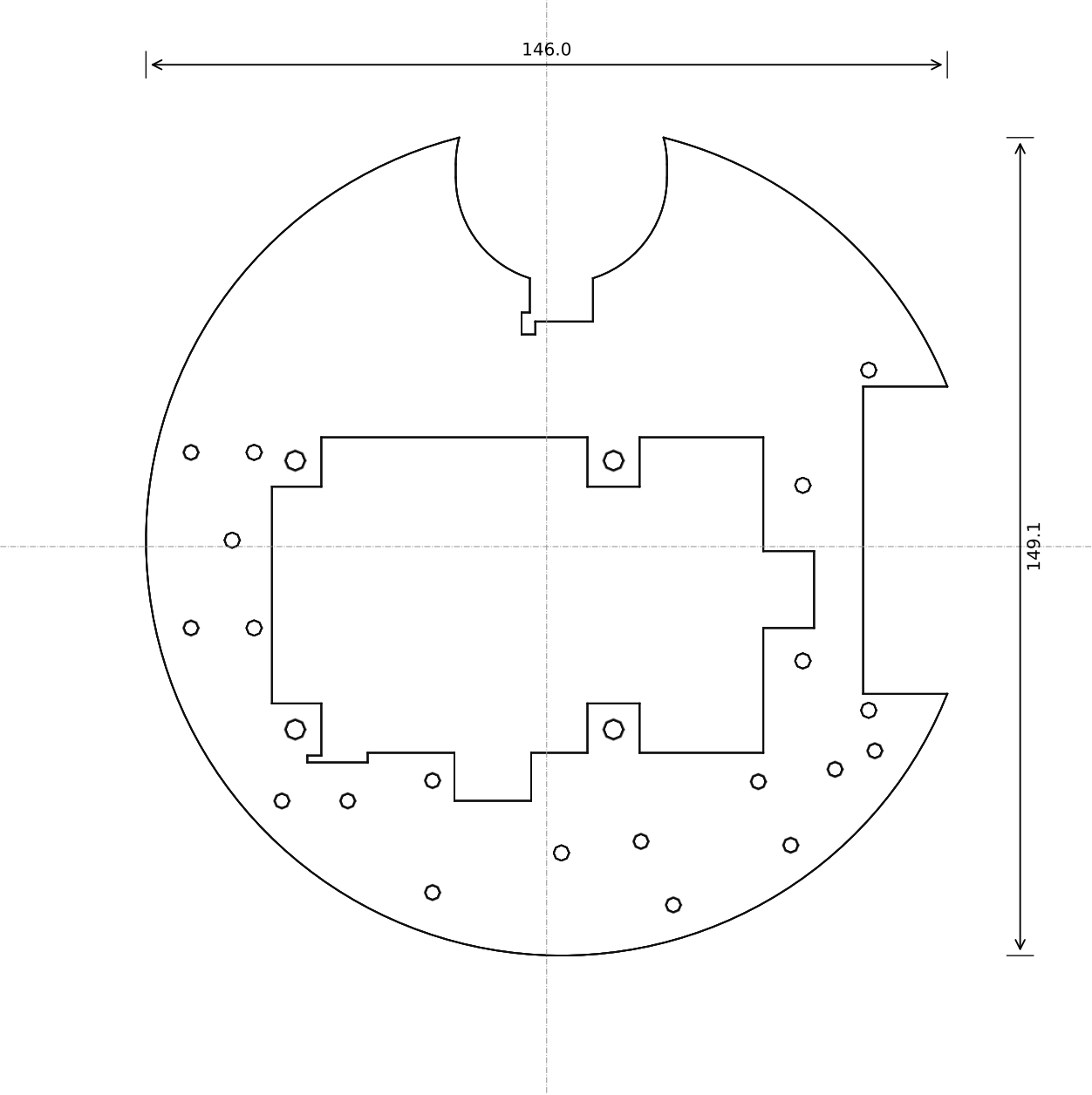
its edges) 0.35 above the openings' chamfered mouths
 min. tolerance ± 0.1 unless stated; clearances: turning 0.30,

Volume 32.24 cm³

static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 43 to 75, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. disc r 75.7 x 1.0, flush in the core's recess on a 0.2 die-cut gasket (two pieces); same Pi window, carriage hole and port slot as the core
- 2. 12 x Ø2.2 countersunk from below (DIN 7991 M2 heads, 0.2 proud) for the boards' studs
- 3. four 9.5 square lands reaching into the Pi window at the Pi's holes [(-48.5, -34.5), (9.5, -34.5), (-48.5, 14.5), (9.5, 14.5)], Ø2.7 csk from below: the Pi and adapter stack screws to them (M2.5 x 6 into the standoffs)
- 4. 9 x M2.5 countersunk; unscrew to clean the fin channels

the 60 — v16

Closing plate (aluminium)

Cadrane · sheet 20 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

Material: 1 mm aluminium, laser-cut, black anodised (outer face)

Process: laser-cut, not printed

mm. Tolerance ± 0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: closing_plate_AL

Overall 146.0 x 149.1 x 1.0

Volume 10.68 cm³

Scale: see each view (fit)